

Digital Design (SET 111)

L07

Boolean Algebra and Logic Simplification

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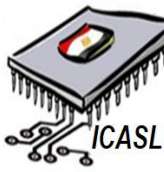


- ☐ **Standard Forms of Boolean Expressions**
- ☐ **Boolean Expressions and Truth Tables**
- ☐ **The Karnaugh Map (K-map)**
- ☐ **Map the standard SOP expression in the K-Map**
- ☐ **K-Map Simplification of SOP Expressions**
- ☐ **K-Map Simplification of Truth Table**
- ☐ **K-Map Simplification of Logic Circuit**



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Standard Forms of Boolean Expressions



□ All Boolean expressions can be converted into either of **two standard** forms.

□ The sum-of-products (**SOP**) form.

Ex.1. $X = \underbrace{AB}_{G1} + \underbrace{BCD}_{G2} + \underbrace{AC}_{G3}$

$\underbrace{\hspace{10em}}_{G4}$

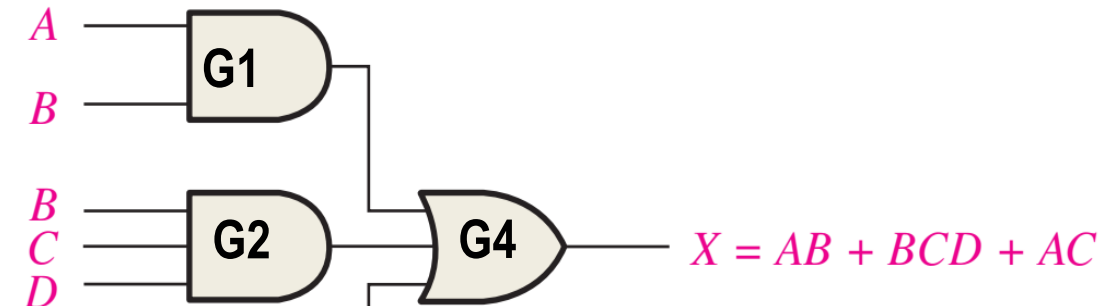
A **SOP** expression is equal to **1**.

□ The product-of-sums (**POS**) form.

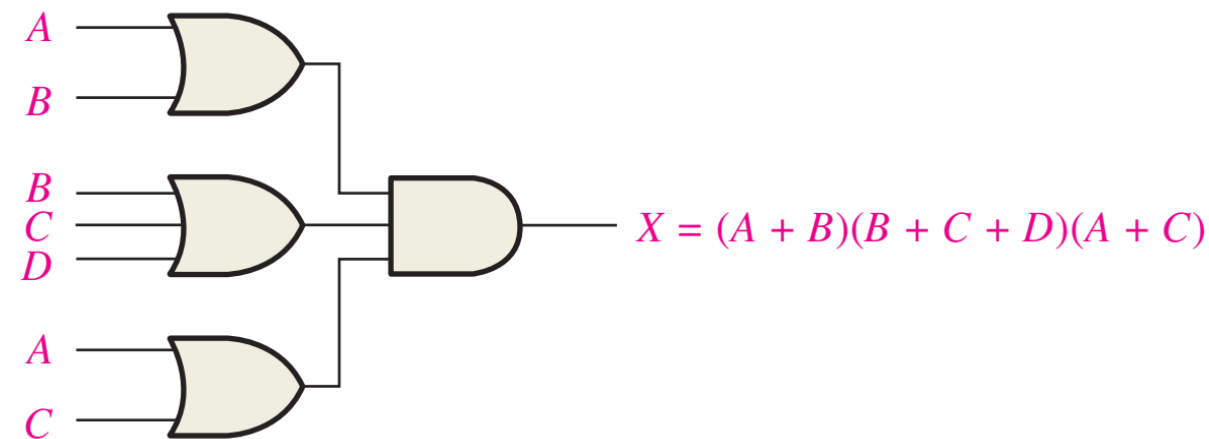
Ex.2. $X = \underbrace{(A + B)}_{G1} \cdot \underbrace{(B + C + D)}_{G2} \cdot \underbrace{(A + C)}_{G3}$

$\underbrace{\hspace{10em}}_{G4}$

A **POS** expression is equal to **0**



Implementation of the **SOP** expression



Implementation of the **POS** expression



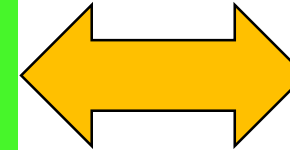
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Boolean Expressions and Truth Tables (1/7)



- ❑ The **truth table** is a common way of presenting the logical operation of a circuit.
- ❑ Standard **SOP** or **POS** expressions can be determined from a truth table and vice versa

$$\begin{aligned} X &= AB + BCD + AC && \text{SOP} \\ X &= (A + B) \cdot (B + C + D) \cdot (A + C) && \text{POS} \end{aligned}$$



Input			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

❑ Remember

- ❑ A truth table is simply a list of the possible combinations of **input** variable values and the corresponding **output** values (**1 or 0**).
- ❑ For an expression with a domain of **two** variables, there are **four** different combinations of those variables ($2^2 = 4$).
- ❑ For an expression with a domain of **three** variables, there are **eight** different combinations of those variables ($2^3 = 8$).

Boolean Expressions and Truth Tables (2/7)



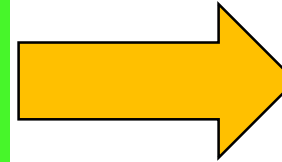
- ❑ The **truth table** is a common way of presenting the logical operation of a circuit.
- ❑ Standard **SOP** or **POS** expressions can be determined from a truth table and vice versa

$$X = AB + BCD + AC$$

SOP

$$X = (A + B). (B + C + D). (A + C)$$

POS



Input			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Boolean Expressions and Truth Tables (3/7)



□ Converting SOP Expressions into Truth Table Format

Ex.1 Develop a truth table for the standard SOP expression

$$X = \overline{A} \overline{B} C + A \overline{B} \overline{C} + A B C$$

Ans.

□ 1st Step: Build the truth table

of variables = 3

of possible combinations = $2^3 = 8$

□ 2nd Step: Fill in the table according to SOP

A SOP expression is equal to 1.

$\overline{A}$	$\overline{B}$	C	A	$\overline{B}$	$\overline{C}$	A	B	C
↓	↓	↓	↓	↓	↓	↓	↓	↓
0	0	1	1	0	0	1	1	1

Input			Output	
A	B	C	X	Product Term
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Boolean Expressions and Truth Tables (4/7)



□ Converting POS Expressions into Truth Table Format

Ex.1 Develop a truth table for the standard POS expression

$$X = (A + B + C)(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + B + \overline{C})(\overline{A} + \overline{B} + C)$$

Ans.

□ 1st Step: Build the truth table

of variables = 3

of possible combinations = $2^3 = 8$

□ 2nd Step: Fill in the table according to POS

A POS expression is equal to 0

$$(A + B + C) (A + \overline{B} + C) (A + \overline{B} + \overline{C}) (\overline{A} + B + \overline{C}) (\overline{A} + \overline{B} + C)$$

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
0 0 0 0 1 0 0 1 1 1 0 1 1 1 0

Input			Output	Sum Term
A	B	C	X	
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Boolean Expressions and Truth Tables (5/7)



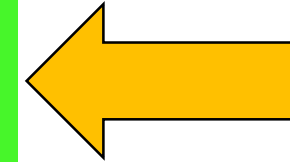
- ❑ The **truth table** is a common way of presenting the logical operation of a circuit.
- ❑ Standard **SOP** or **POS** expressions can be determined from a truth table and vice versa

$$X = AB + BCD + AC$$

SOP

$$X = (A + B). (B + C + D). (A + C)$$

POS



Input			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Boolean Expressions and Truth Tables (6/7)



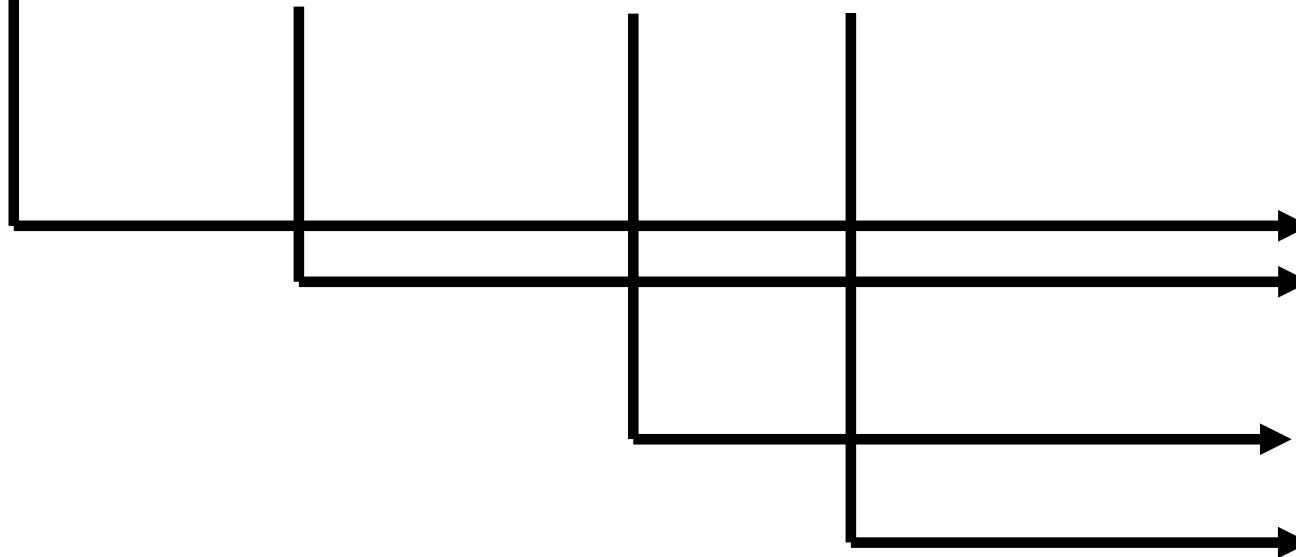
□ Converting Truth Table Format into Standard Expressions

Ex.1 Determine the standard **SOP** expression and the equivalent standard **POS** expression.

Ans.

□ To get SOP → 1

$$X = \overline{A} B C + A \overline{B} \overline{C} + A B \overline{C} + A B C$$



Input			Output
A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Boolean Expressions and Truth Tables (7/7)



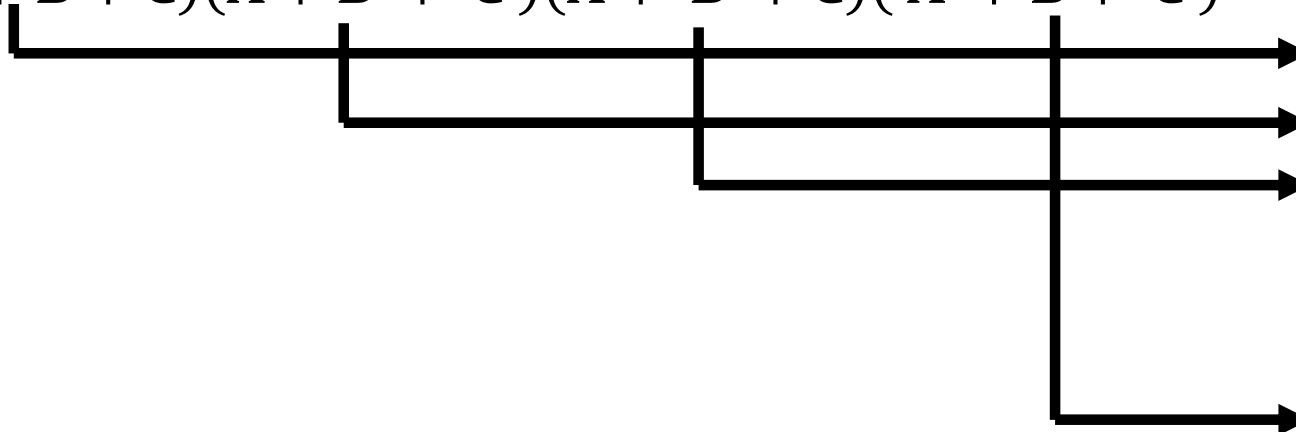
□ Converting Truth Table Format into Standard Expressions

Ex.1 Determine the standard **SOP** expression and the equivalent standard **POS** expression.

Ans.

□ To get POS $\rightarrow 0$

$$X = (A + B + C)(A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + \overline{C})$$



Input			Output
A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

SOP and the POS expressions derived in this example are equivalent.





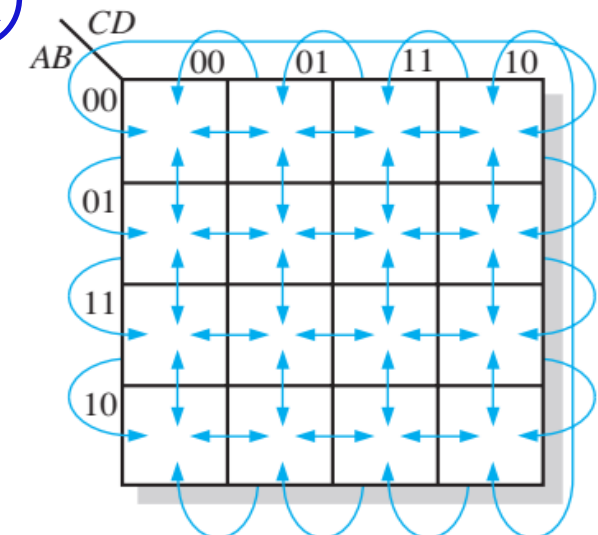
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The Karnaugh Map



❑ Definition

- ❑ A **Karnaugh** map provides a systematic method for simplifying Boolean expressions. It provides a “**cookbook**” method for simplification. 😊
- ❑ It is an array of cells in which each cell represents a binary value of the input variables.
- ❑ **Notes:** there is only a single-variable change between adjacent cells. 😊
- ❑ The number of cells in a Karnaugh map is **equal** to the total number of possible input variable combinations= the number of rows in a truth table. 😊
- ❑ Ex.1 2-Variable needs $2^2 = 4$ cells.
- ❑ Ex.1 3-variables needs $2^3 = 8$ cells.
- ❑ Ex. 2 Four variables needs $2^4 = 16$ cells.

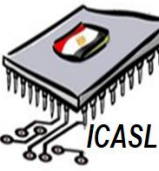


Adjacent cells on a K-Map



- ☐ Standard Forms of Boolean Expressions
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Map the standard SOP expression in the K-Map (1/5)



- Map the standard SOP expression in the K-Map
 - The 2-variables Karnaugh Map

Ex.1

$$F = A \cdot \overline{B} + \overline{A} \cdot B$$

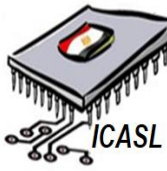
		B	
		0	1
A	0	0	1
	1	1	0

Ex.2

$$F = \overline{A} \cdot \overline{B} + A \cdot B$$

		B	
		0	1
A	0	1	0
	1	0	1

Map the standard SOP expression in the K-Map (2/5)



- Map the standard SOP expression in the K-Map
 - The 3-variables Karnaugh Map

Ex.1

$$F = A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot \overline{B} \cdot \overline{C} + A \cdot \overline{B} \cdot C + \overline{A} \cdot B + B \cdot \overline{C}$$

		B			
		00	01	11	10
A	0	1	0	1	1
	1	1	1	0	1

Ex.2

$$F = \overline{A} \cdot \overline{B} \cdot C + \overline{A} \cdot B \cdot \overline{C} + A \cdot B \cdot \overline{C} + ABC$$

		B			
		00	01	11	10
A	0	0	1	0	1
	1	0	0	1	1

Map the standard SOP expression in the K-Map (3/5)



- Map the standard SOP expression in the K-Map
 - The 3-variables Karnaugh Map

Ex.3

$$F = \overline{A} . B . C + A . \overline{B} . C + A . \overline{B} . \overline{C}$$

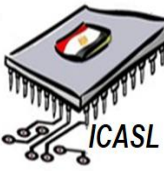
		B			
		00	01	11	10
A	0	0	0	1	0
	1	1	1	0	0

Ex.4

$$F = \overline{A} + A . \overline{B} + A . B . \overline{C}$$

		B			
		00	01	11	10
A	0	1	1	1	1
	1	1	1	0	1

Map the standard SOP expression in the K-Map (4/5)



- Map the standard SOP expression in the K-Map
 - The 4-variables Karnaugh Map

Ex.1

$$F = \overline{A} \cdot \overline{C} \cdot \overline{D} + A \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot C \cdot \overline{D} + A \cdot C \cdot \overline{D}$$

		C			
		00	01	11	10
A	00	1	1	0	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	0	0	1
		D			

Ex.2

$$F = \overline{B} \cdot \overline{C} + A \cdot \overline{B} + A \cdot B \cdot \overline{C} + A \cdot \overline{B} \cdot C \cdot \overline{D} + \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot D + A \cdot \overline{B} \cdot C \cdot D$$

		C			
		00	01	11	10
A	00	1	1	0	0
	01	0	0	0	0
	11	1	1	0	0
	10	1	1	1	1
		D			

Map the standard SOP expression in the K-Map (5/5)



- Map the standard SOP expression in the K-Map
 - The 4-variables Karnaugh Map

Ex.3

$$F = A + \overline{C} \cdot D + A \cdot C \cdot \overline{D} + \overline{A} \cdot B \cdot C \cdot \overline{D}$$

$\begin{matrix} \text{CD} \\ \text{AB} \end{matrix}$		$\overbrace{\quad\quad\quad}^C$			
		00	01	11	10
$\underbrace{\quad\quad\quad}_A$	00	0	1	0	0
	01	0	1	0	1
	11	1	1	1	1
	10	1	1	1	1
		$\underbrace{\quad\quad\quad}_D$			

Ex.4

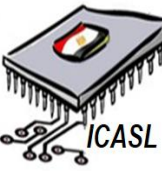
$$F = \overline{B} \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot B \cdot \overline{C} \cdot \overline{D} + A \cdot B \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot \overline{B} \cdot C \cdot D + A \cdot \overline{B} \cdot C \cdot D + \overline{A} \cdot \overline{B} \cdot C \cdot \overline{D} + \overline{A} \cdot B \cdot C \cdot \overline{D} + A \cdot B \cdot C \cdot \overline{D} + A \cdot \overline{B} \cdot C \cdot \overline{D}$$

$\begin{matrix} \text{CD} \\ \text{AB} \end{matrix}$		$\overbrace{\quad\quad\quad}^C$			
		00	01	11	10
$\underbrace{\quad\quad\quad}_A$	00	1	0	1	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	0	1	1
		$\underbrace{\quad\quad\quad}_D$			



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K-Map Simplification of SOP Expressions (1/6)



- ☐ ***Fewest*** possible terms with the ***Fewest*** possible variables. 😊
- ☐ A minimum **SOP** expression is obtained by grouping the **1s**
 - ☐ A group must contain either 1, 2, 4, 8, or 16 cells, which are all powers of **two**. In the case of a 3-variable map, $2^3 = 8$ cells is the maximum group.
 - ☐ **Always** include the **largest** possible number of 1s in a group .
 - ☐ Each 1 on the map must be included in at **least** one group. The 1s already in a group can be **included** in another group as long as the overlapping groups include non-common 1s
 - ☐ Each group of cells containing 1s creates one **product term**.
 - ☐ When all the minimum product terms are derived from the **Karnaugh** map, they are summed to form the minimum **SOP** expression.

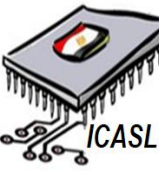
For a 3-variable map

- (1) A 1-cell group yields a 3-variable product term
- (2) A 2-cell group yields a 2-variable product term
- (3) A 4-cell group yields a 1-variable term
- (4) An 8-cell group yields a value of 1 for the expression.

For a 4-variable map

- (1) A 1-cell group yields a 4-variable product term
- (2) A 2-cell group yields a 3-variable product term
- (3) A 4-cell group yields a 2-variable product term
- (4) An 8-cell group yields a 1-variable term
- (5) A 16-cell group yields a value of 1 for the expression

K-Map Simplification of SOP Expressions (2/6)



□ The 2-variables Karnaugh Map

Ex.1

$$F = A \cdot \overline{B} + \overline{A} \cdot B + A \cdot B$$



Ex.1

$$F = A + B$$

		B	
		0	1
A	0	0	1
	1	1	1

Ex.2

$$F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{A} \cdot \overline{B}$$

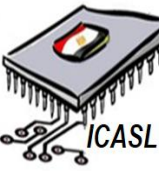


Ex.2

$$F = \overline{A} + \overline{B}$$

		B	
		0	1
A	0	1	1
	1	1	0

K-Map Simplification of SOP Expressions (3/6)



❑ The 3-variables Karnaugh Map

Ex.1

$$F = \overline{A} . \overline{B} . C + \overline{A} . B . \overline{C} + A . B . \overline{C} + ABC$$



Ex.1

$$F = A . B + B . \overline{C} + \overline{A} . \overline{B} . C$$

A \ BC	00	01	11	10
0	0	1	0	1
1	0	0	1	1

Ex.1

$$F = A . \overline{B} . \overline{C} + \overline{A} . \overline{B} . \overline{C} + A . \overline{B} . C + \overline{A} . B + B . \overline{C}$$



Ex.1

$$F = \overline{C} + \overline{A} . B + A . \overline{B}$$

A \ BC	00	01	11	10
0	1	0	1	1
1	1	1	0	1

K-Map Simplification of SOP Expressions (4/6)



❑ The 3-variables Karnaugh Map

Ex.3

$$F = \overline{A} . B . C + A . \overline{B} . C + A . \overline{B} . \overline{C}$$



Ex.3

$$F = A . \overline{B} + \overline{A} . B . C$$

A \ BC	B			
	00	01	11	10
0	0	0	1	0
1	1	1	0	0

Ex.4

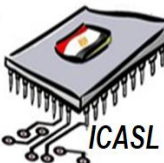
$$F = \overline{A} + A . \overline{B} + A . B . \overline{C}$$



Ex.4

$$F = \overline{A} + \overline{B} + \overline{C}$$

A \ BC	B			
	00	01	11	10
0	1	1	1	1
1	1	1	0	1



K-Map Simplification of SOP Expressions (5/6)

❑ The 4-variables Karnaugh Map

Ex.1

$$F = \overline{A} \cdot \overline{C} \cdot \overline{D} + A \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot C \cdot \overline{D} + A \cdot C \cdot \overline{D}$$



Ex.1

$$F = \overline{D} + \overline{A} \cdot \overline{B} \cdot \overline{C}$$

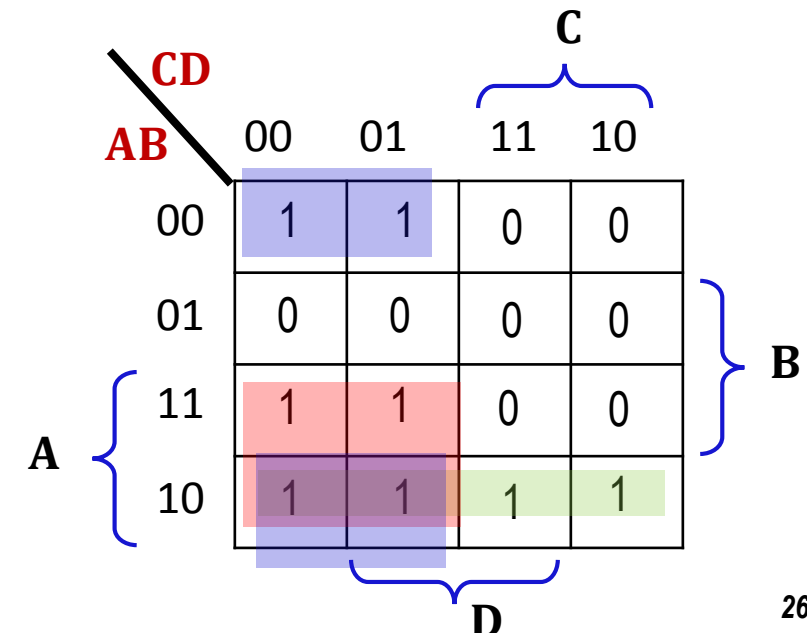
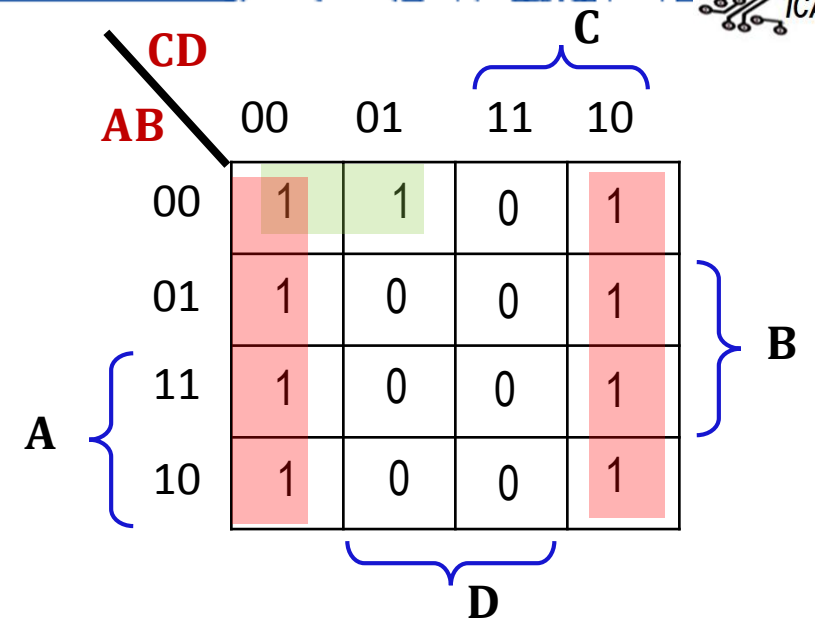


Ex.2

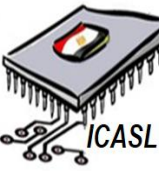
$$F = \overline{B} \cdot \overline{C} + A \cdot \overline{B} + A \cdot B \cdot \overline{C} + A \cdot \overline{B} \cdot C \cdot \overline{D} + \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot D + A \cdot \overline{B} \cdot C \cdot D$$

Ex.2

$$F = A \cdot \overline{B} + A \cdot \overline{C} + \overline{B} \cdot \overline{C}$$



K-Map Simplification of SOP Expressions (6/6)



❑ The 4-variables Karnaugh Map

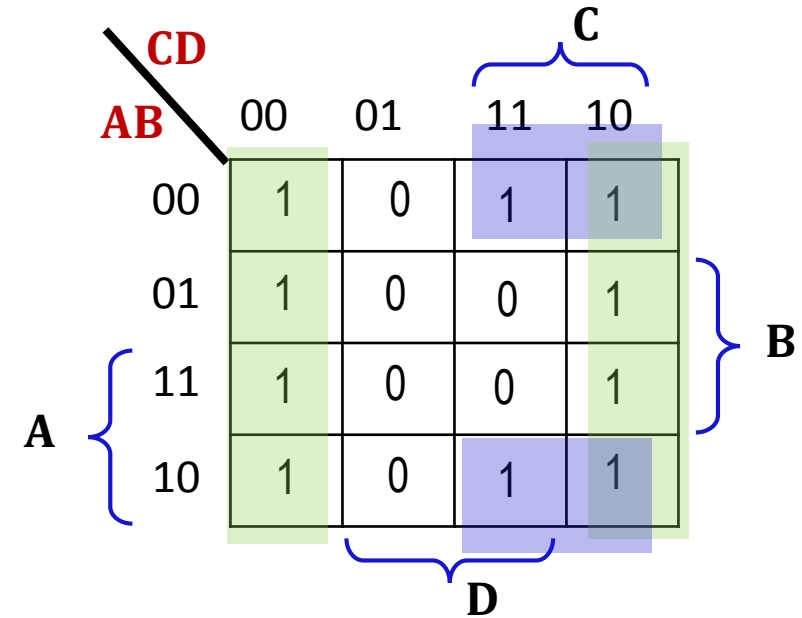
Ex.3

$$F = \overline{B} \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot B \cdot \overline{C} \cdot \overline{D} + A \cdot B \cdot \overline{C} \cdot \overline{D} \\ + \overline{A} \cdot \overline{B} \cdot C \cdot D + A \cdot \overline{B} \cdot C \cdot D + \overline{A} \cdot \overline{B} \cdot C \cdot \overline{D} + \overline{A} \cdot B \cdot C \cdot \overline{D} \\ + A \cdot B \cdot C \cdot \overline{D} + A \cdot \overline{B} \cdot C \cdot \overline{D}$$



Ex.3

$$F = \overline{D} + \overline{B} \cdot C$$





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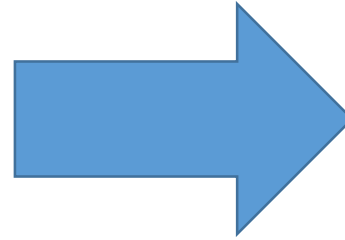
K-Map Simplification of Truth Table (1/3)



□ The 2-variables Karnaugh Map

Truth Table

Input		Output
A	B	F
0	0	0
0	1	1
1	0	1
1	1	0



		B	
		0	1
A	0	m0	m1
	1	m2	m3

		B	
		0	1
A	0	0	1
	1	1	0

$$F = A \cdot \overline{B} + \overline{A} \cdot B$$

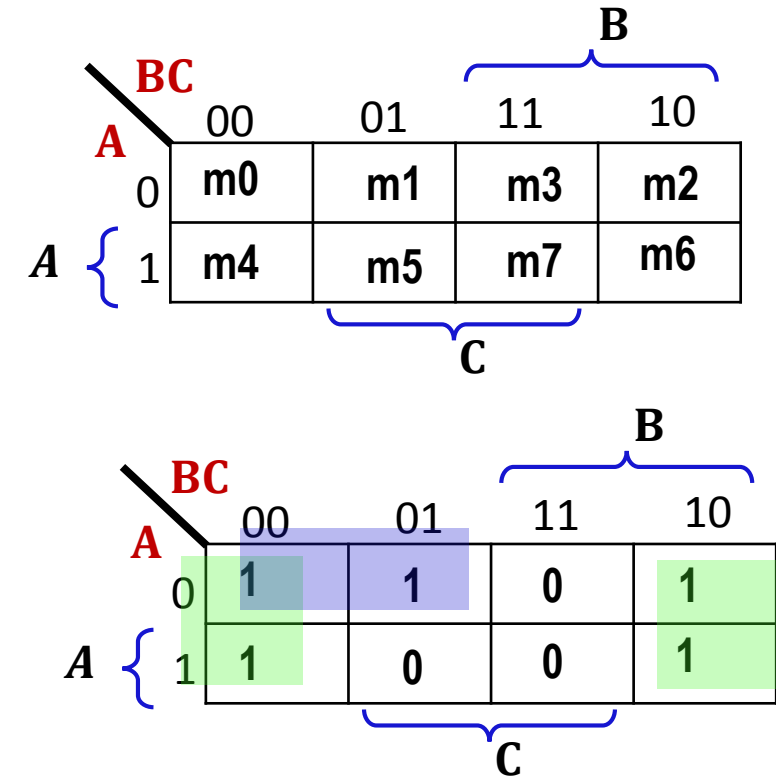
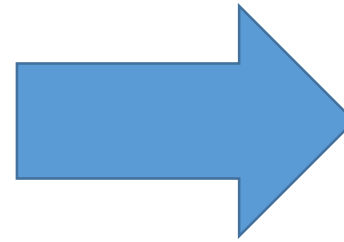
K-Map Simplification of Truth Table (2/3)



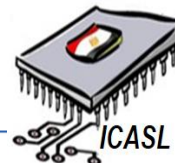
❑ The 3-variables Karnaugh Map

Truth Table

Input			Output
A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



$$F = \overline{C} + \overline{A} \cdot \overline{B}$$

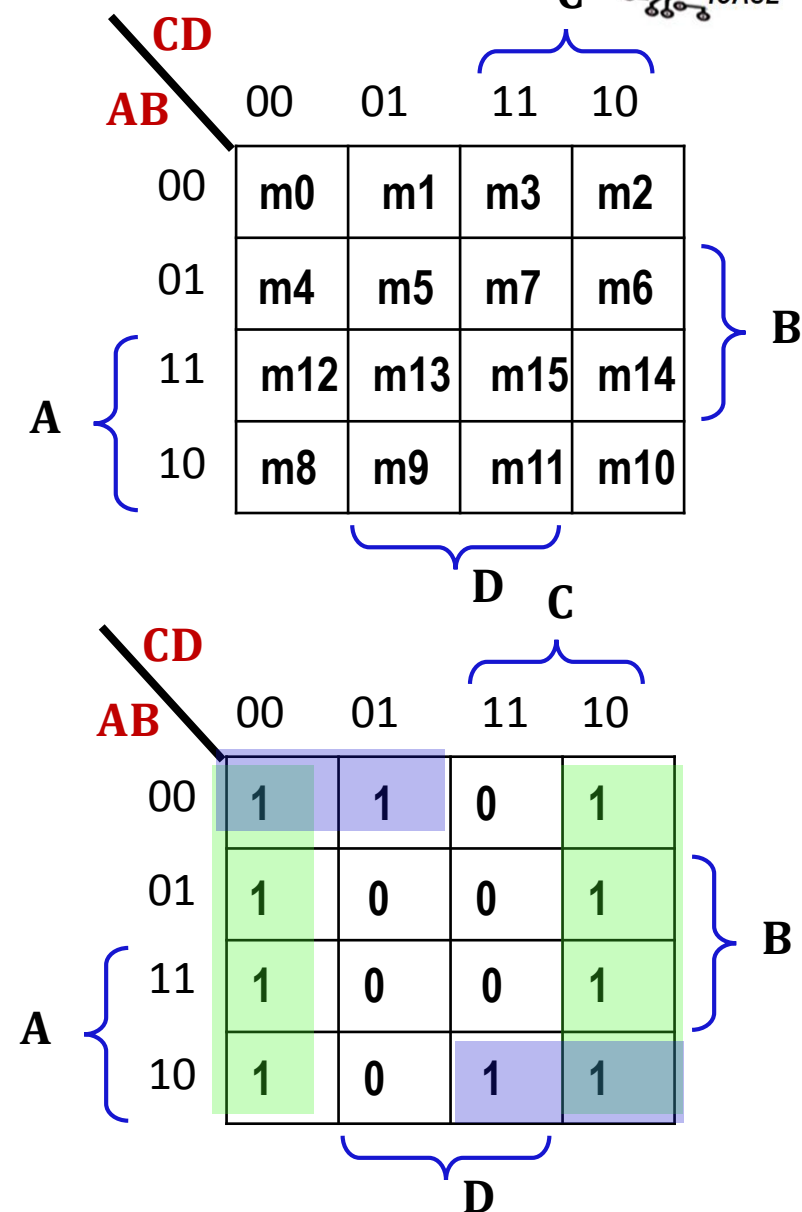
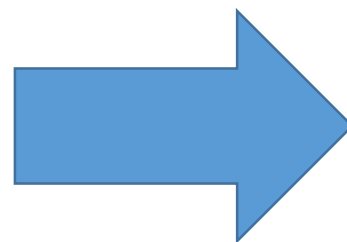


K-Map Simplification of Truth Table (3/3)

❑ The 4-variables Karnaugh Map

Truth Table

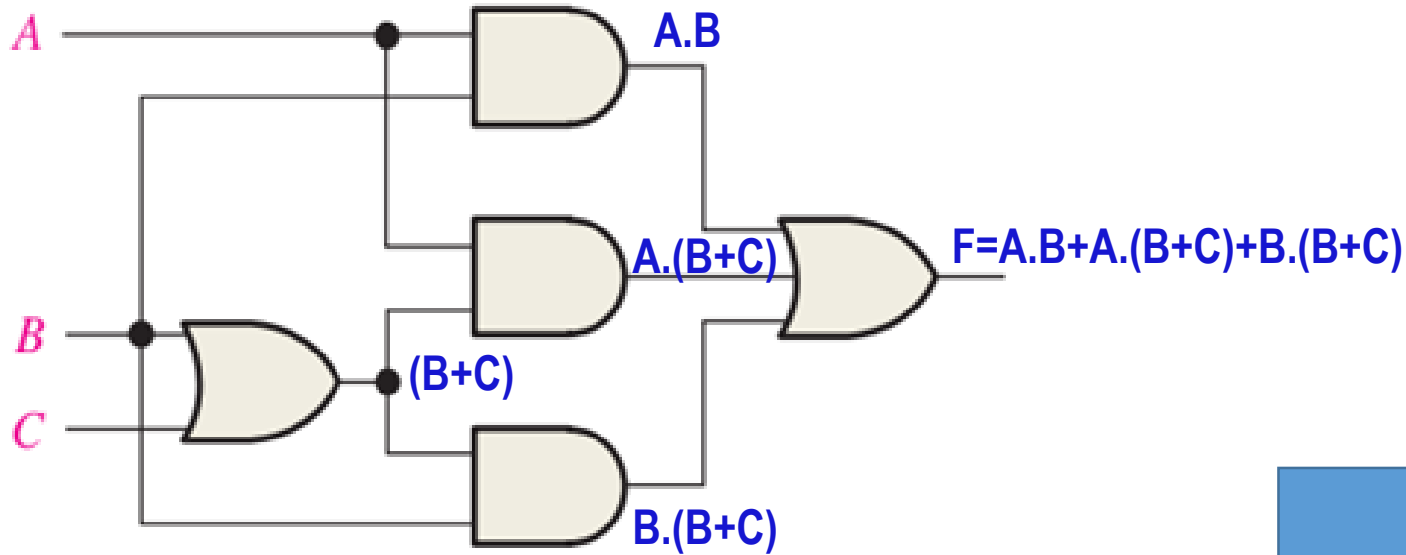
Input				Output
A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0





- ☐ Standard Forms of Boolean Expressions
- ☐ Boolean Expressions and Truth Tables
- ☐ The Karnaugh Map (K-map)
- ☐ Map the standard SOP expression in the K-Map
- ☐ K-Map Simplification of SOP Expressions
- ☐ K-Map Simplification of Truth Table
- ☐ **K-Map Simplification of Logic Circuit**

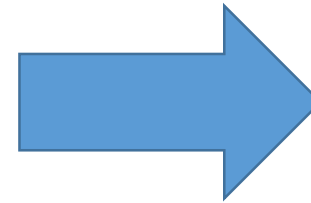
K-Map Simplification of Logic Circuit (4/4)



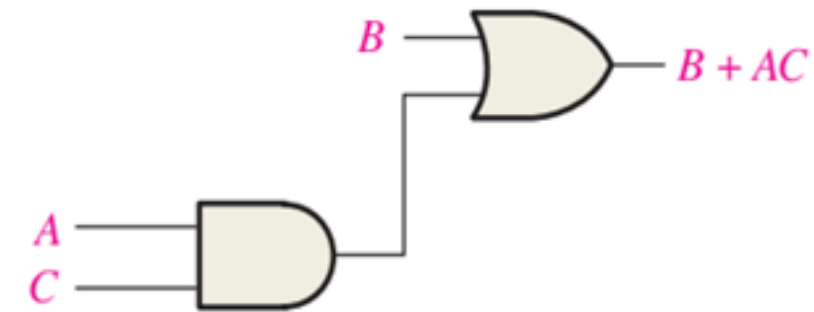
$$F = A.B + A.B + A.C + B.B + B.C$$

$$F = A.B + A.C + B + B.C$$

$$F = B + AC$$



		B			
		C		11	10
A	0	0	0	1	1
	1	0	1	1	1



Thanks!